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Capstone Project Report
on
PLC-Based Preventive Maintenance Warning System

Submitted in partial fulfillment of the requirements
of the Third Year of Bachelor of Technology
in Electronics and Computer Science

by

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Certificate of Approval

This is to certify that the project report entitled "PLC-Based Preventive Maintenance Warning System" is submitted by the students named below in partial fulfillment of the requirements for the course Programmable Logic Controllers and SCADA in the Department of Electronics and Computer Science, **St. John College of Engineering and Management, (An autonomous college affiliated to the University of Mumbai.)**

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I/we declare that this written submission represents my/our ideas in my/our own words and where others ideas or words have been included, I/we have adequately cited and referenced the original sources. I/we also declare that I/we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my/our submission. I/we understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Preventive maintenance reduces avoidable equipment stoppages by ensuring that routine service activities are completed at defined intervals. This project presents a PLC-based warning system for four maintenance schedules: weekly lubrication and cleaning, monthly hydraulic-oil checking and filter replacement, quarterly pneumatic-system inspection, and yearly sensor calibration. The proposed control is organized around a Mitsubishi FX5U programmable logic controller programmed in MELSOFT GX Works3. Four interval-completion conditions are represented by timer or time-base flags T0-T3. When an interval becomes due, its corresponding warning output Y0-Y3 is latched until the associated reset input is operated after the maintenance activity is completed. A common buzzer output Y4 is energized whenever any one of the four warning outputs is active. This arrangement separates maintenance tasks, keeps the warning visible until acknowledgement, and provides a single audible indication for operators and technicians. The report includes the operating concept, functional requirements, I/O allocation, process flow, ladder-diagram evidence and a logic-operation table. The supplied GX Works3 screenshot supports discussion of the ladder-monitoring environment; it does not, by itself, prove that week-, month-, quarter- or year-long intervals have elapsed in a physical installation. For practical deployment, the long intervals should be implemented with a suitable real-time clock/calendar comparison or cascaded counters, and the timer ranges, input wiring, reset policy and output ratings must be verified against the selected FX5U hardware and plant requirements.

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Note: Student names, roll numbers and guide details are left as editable placeholders for final submission.

CHAPTER 1: OVERVIEW

1.1 Introduction

Preventive maintenance is a planned activity intended to keep equipment in an acceptable operating condition and reduce failures caused by missed service tasks. The proposed system uses PLC ladder logic to track four recurring maintenance intervals and to alert personnel when a task becomes due. It is intended for plant equipment that requires regular lubrication, cleaning, hydraulic checks, filter changes, pneumatic inspection and sensor calibration.

The Mitsubishi FX5U PLC provides the sequential control platform. MELSOFT GX Works3 is used to organize and monitor the ladder program. Each maintenance category has a separate warning output, while a common buzzer alerts staff whenever any category is overdue.

1.2 Background and Need

In a manual process, maintenance dates may be recorded in notebooks, spreadsheets or shift checklists. Busy production schedules, staff changes and incomplete handovers can cause a task to be delayed or forgotten. A local PLC warning does not replace the maintenance procedure, but it provides a consistent reminder at the machine and allows warning acknowledgement to be handled in the control logic.

1.3 Importance of the Project

- Separate schedules: weekly, monthly, quarterly and yearly tasks are tracked independently.
- Persistent warning: a due warning remains active until the corresponding reset is operated.
- Common audible alert: output Y4 is driven by the logical OR of Y0, Y1, Y2 and Y3.
- Simple implementation: timers/time-base flags, reset inputs and output coils can be represented in standard ladder logic.
- Expandable design: a future HMI or SCADA screen can display due dates, task history and acknowledgements.

1.4 Maintenance Categories

Frequency	Maintenance activity	Purpose
Weekly	Lubrication and cleaning	Reduce friction, remove deposits and keep moving parts accessible for inspection.
Monthly	Hydraulic oil check and filter change	Check oil condition/level and renew filters according to the equipment manual.
Quarterly	Pneumatics check	Inspect air leakage, pressure, tubing, regulators, valves and FRL units.
Yearly	Sensor calibration	Verify sensor readings against a suitable reference and recalibrate as required.

CHAPTER 2: LITERATURE SURVEY & PROPOSED WORK

2.1 Introduction and Comparative Survey

Maintenance reminders can be implemented in several ways. The table compares common approaches at a system level and identifies why a PLC-based design is suitable where local industrial I/O and a persistent warning are required. The comparison is descriptive; it is not presented as the result of a controlled product benchmark.

Existing approach	Advantages	Limitations / gap
Paper checklist or wall calendar	Low cost; easy to introduce.	Depends on manual checking, handover quality and timely follow-up; no automatic audible alert.
Standalone interval timer	Simple and inexpensive; useful for one periodic task.	Several independent timers may be needed; little interlocking or shared alarm logic.
HMI / SCADA maintenance schedule	Can display asset status, history and reminders centrally.	May require extra hardware/software and configuration for a small standalone machine.
PLC ladder-based warning (proposed)	Uses discrete I/O, independent warning bits and a common buzzer; readily integrated with a machine control panel.	Long interval handling and reset authorization must be engineered and verified for the selected PLC.

2.2 Problem Definition

The problem addressed is the lack of a simple, repeatable mechanism to remind operators about equipment service at multiple frequencies. The system must identify which maintenance category has reached its due interval, retain the warning until the task is acknowledged, and provide a common audible indication without clearing unrelated warnings.

- Missed schedule: a service action may be delayed when reminders are only kept in a paper log or personal calendar.
- Ambiguous status: a single lamp or buzzer cannot identify which maintenance category is due unless each category has its own warning bit/indicator.
- Unintended reset: a single global reset could erase other outstanding maintenance warnings.
- Long timing periods: weekly-to-yearly intervals exceed the range of many ordinary PLC timer instructions when treated as a single preset.

The design response is to maintain separate interval-completion flags and warning outputs, use one reset input per maintenance category, and combine the warning bits only at the common buzzer rung.

2.3 Feasibility Study

Area	Feasibility assessment
Technical	The FX5U ladder environment supports discrete input contacts, output coils, timer/counter logic and internal flags. An RTC/calendar comparison or cascaded counter structure can represent multi-day intervals. Input and output wiring must match the selected PLC model.
Operational	Four separate warnings align with the maintenance schedule. Technicians can identify the required task and reset only the relevant warning after the work is completed.
Economic	The concept can use the PLC and panel indicators already available in a machine. For an existing installation, the incremental cost may be limited to pushbuttons, a buzzer, indicators and wiring; actual cost depends on the installation.
Safety and reliability	The buzzer is an advisory device, not a safety-rated function. The warning system must not bypass guards, emergency stops or OEM maintenance instructions.

2.4 Proposed Methodology

1. Requirement analysis: list the four maintenance activities, intervals and responsible staff.
2. I/O allocation: assign X0-X3 to task-specific reset inputs; allocate T0-T3 or equivalent time-base flags to the four intervals; map Y0-Y3 to the warning states and Y4 to the common buzzer.
3. Interval tracking: run each time base continuously while its reset input is inactive. Use a suitable RTC/calendar comparison or cascaded counters for extended timing.
4. Warning latch: when a time base reaches its preset, energize the corresponding warning output and retain it through a seal-in branch.
5. Reset and restart: after performing the maintenance task, the technician presses that task's reset input. Only the associated warning is cleared and its interval is restarted.
6. Common annunciation: combine Y0-Y3 in parallel contacts so that any active warning energizes Y4.
7. Verification: test each warning independently, each reset independently, simultaneous warnings, and the stop/start behavior before connecting field devices.

Demonstration intervals: 7 days (weekly), 30 days (monthly), 90 days (quarterly) and 365 days (yearly). For plant use, follow the equipment OEM schedule and a calendar-based policy where month-end or leap-year accuracy matters.

CHAPTER 3: ANALYSIS AND PLANNING

3.1 System Requirements

Requirement type	Requirement
Functional	Track four independent maintenance intervals and detect when each is due.
Functional	Hold the task-specific warning until the matching reset is activated.
Functional	Energize a common buzzer if one or more warning outputs are active.
Functional	Ensure that resetting one task does not clear another active task warning.
Non-functional	Operate locally without depending on internet connectivity.
Non-functional	Use clear tag comments, documented timer bases and protected wiring in the control panel.
Non-functional	Retain correct elapsed-time behavior across PLC power interruption when required by the plant policy.

3.2 Project Planning

Phase	Activity	Deliverable	Duration
1	Study maintenance requirements and existing control environment.	Problem statement and task schedule.	Week 1
2	Identify PLC I/O, time base and warning/reset behavior.	I/O list and logic specification.	Week 2
3	Develop timer, warning-latch and buzzer rungs in GX Works3.	Ladder program with device comments.	Weeks 3-4
4	Simulate individual and concurrent warning conditions.	Test record and corrected program.	Week 5
5	Check output hardware, field wiring and reset procedure.	Panel-level verification checklist.	Week 6
6	Document results, constraints and future expansion.	Final report and handover notes.	Week 7

3.3 Device Allocation (Timer Concept)

Device	Type	Function
SM400	Special relay	PLC RUN / always-ON condition for timer rung, where supported by the chosen FX5U project.
X0-X3	Digital inputs	Weekly, monthly, quarterly and yearly reset inputs, respectively.
T0-T3	Timer / time-base flags	Interval completion: weekly, monthly, quarterly and yearly.
Y0-Y3	Digital outputs	Weekly, monthly, quarterly and yearly warning outputs.
Y4	Digital output	Common maintenance buzzer, energized by any active warning.

Address consistency note: the GX Works3 monitor screenshot supplied with the report shows paired inputs X0-X7 driving Y0-Y3, while the separate timer-diagram reference uses X0-X3 as reset inputs for T0-T3. The intended function is consistent, but the actual download project must use one reconciled I/O map and matching device comments.

CHAPTER 4: DESIGN & IMPLEMENTATION

4.1 Data Flow Diagram (DFD)

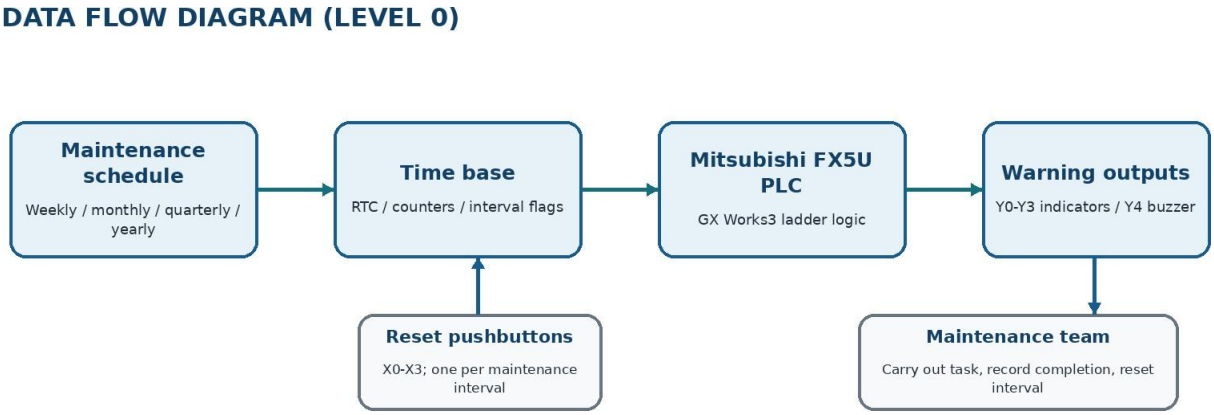


Figure 1: Data flow from the maintenance schedule and reset inputs through the PLC to warning outputs.

4.2 System Block Diagram

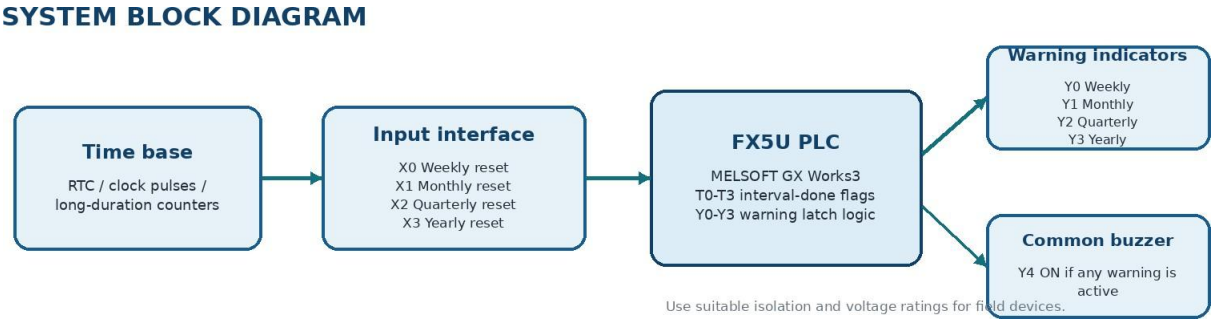


Figure 2: Proposed PLC hardware and ladder-logic interface for four interval warnings.

The time base provides elapsed-time or calendar-comparison information. Reset inputs acknowledge maintenance completion, while output coils drive the task indicators and common buzzer through suitably rated interface circuits.

4.3 Flow Chart and Operating Sequence

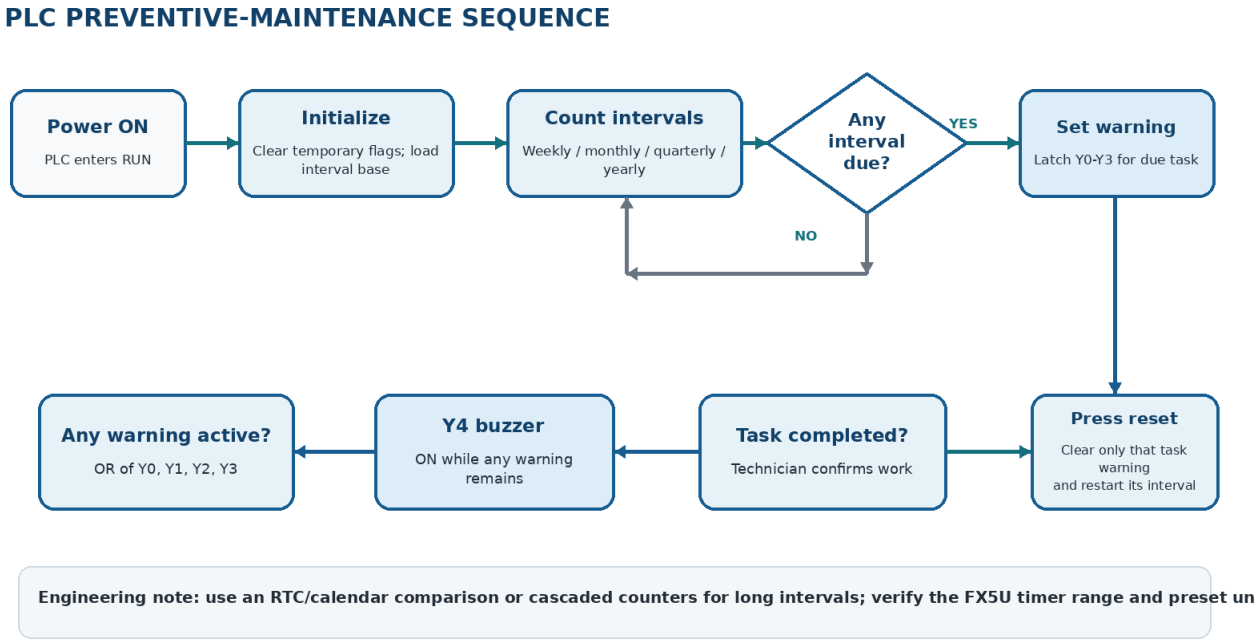


Figure 3: High-level operation of the preventive maintenance warning program.

Sequence Description

8. At PLC start-up, initialize the required flags and confirm that the time base is valid.
9. Continuously evaluate the four maintenance intervals independently.
10. When an interval becomes due, set and hold that category's warning output.
11. If any warning is active, turn ON the common buzzer output Y4.
12. After the maintenance work is completed, operate only the matching reset input. Clear that warning and start a new interval for that task.

Reset is intended to represent maintenance completion, not merely silencing an inconvenient alarm. A production version should define who may reset a warning and how the completed task is recorded.

4.4 Ladder Diagram and GX Works3 Monitoring

The following figures are reproduced from the supplied project images. The first is the logical overview of weekly, monthly, quarterly and yearly warnings. The second is the MELSOFT GX Works3 monitoring screenshot provided for the ladder implementation.

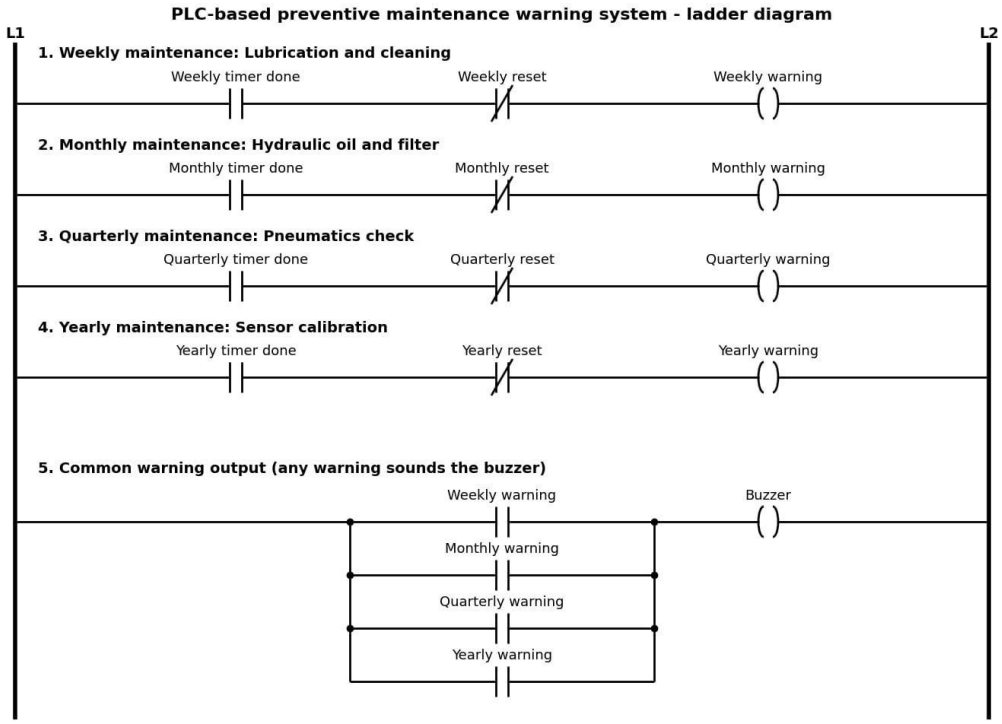


Figure 4: PLC-based preventive maintenance warning system - logical ladder overview.

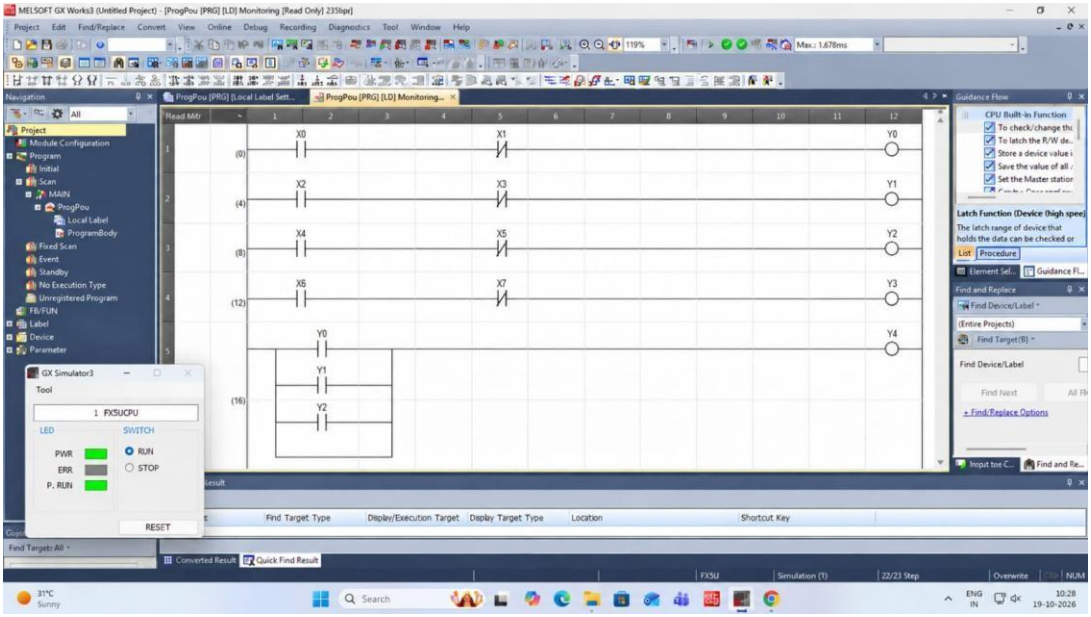


Figure 5: MELSOFT GX Works3 ladder monitoring view with FX5U simulator status.

The screenshots are design evidence. A green RUN / P.RUN indication demonstrates the simulator or CPU status shown in the screenshot; it does not alone demonstrate elapsed maintenance intervals or long-term field performance.

4.5 Detailed Ladder Logic: Timer and Warning Rungs

This panel shows the upper part of the supplied detailed ladder illustration. Each maintenance category uses an interval-done contact and a seal-in branch around its warning output. The corresponding normally closed reset contact breaks the rung when that task is reset.

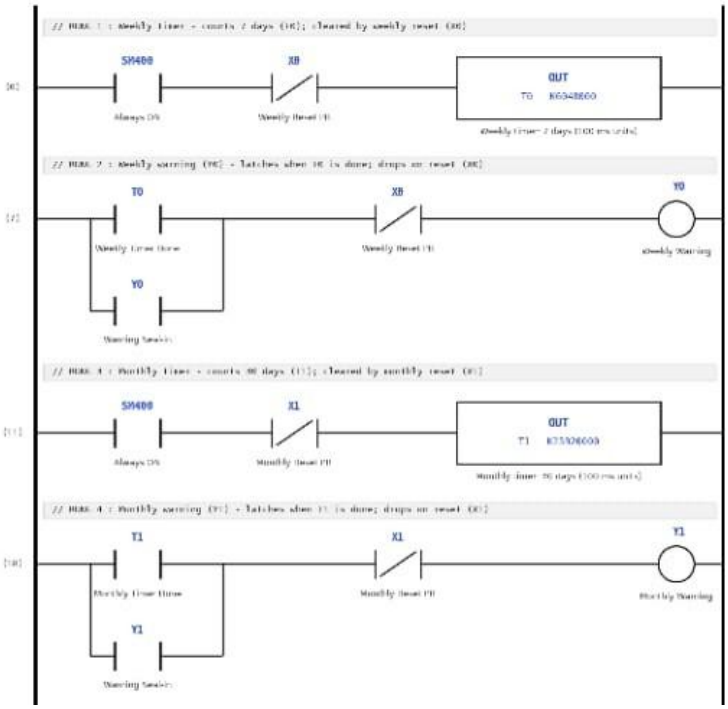


Figure 6: Detailed timer and warning rungs (weekly and monthly timer/warning rungs) from the supplied ladder image.

Device / Address Reference

Device	Assigned function	Logic role
X0	Weekly reset	Clears Y0 and restarts weekly interval.
X1	Monthly reset	Clears Y1 and restarts monthly interval.
X2	Quarterly reset	Clears Y2 and restarts quarterly interval.
X3	Yearly reset	Clears Y3 and restarts yearly interval.
T0-T3	Interval-done flags	Become active after the corresponding time base expires.
Y0-Y3	Task warnings	Remain active until the matching reset is operated.
Y4	Maintenance buzzer	Common audible output for any active warning.

Note: device comments in the actual GX Works3 project should be reconciled with the timer illustration before download; the provided monitor screenshot uses a different paired-input arrangement.

4.6 Common Warning Output and Logic Table

The lower panel of the supplied ladder image shows the yearly-warning rung and the shared maintenance buzzer rung. In the intended common-warning logic, Y0, Y1, Y2 and Y3 are placed in parallel; therefore, any one active maintenance warning energizes Y4.

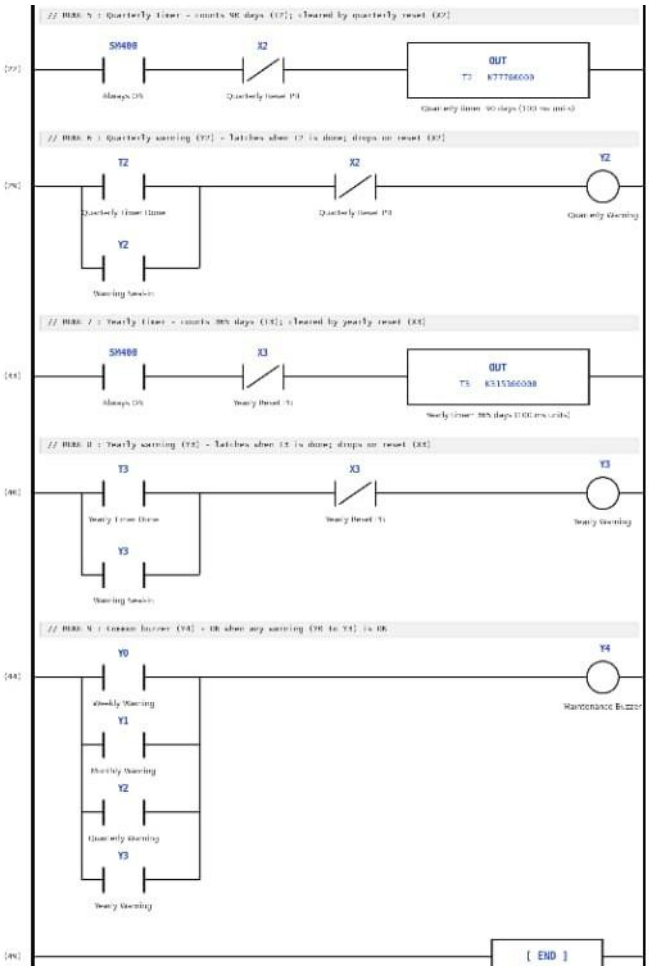


Figure 7: Detailed quarterly/yearly warning and common buzzer rungs.

Logic Operation Table

Condition	Warning output	Y4 buzzer	Expected result
All intervals still counting	Y0-Y3 OFF	OFF	No task is due.
Weekly interval expires	Y0 ON	ON	Weekly lubrication/cleaning reminder appears.
Monthly interval expires	Y1 ON	ON	Hydraulic-oil/filter reminder appears.
Quarterly interval expires	Y2 ON	ON	Pneumatics inspection reminder appears.
Yearly interval expires	Y3 ON	ON	Sensor-calibration reminder appears.
One matching reset is pressed after work	Only that task clears	ON if another warning remains	Other warning states are unaffected.
All active warnings are cleared	Y0-Y3 OFF	OFF	New intervals continue from reset points.

Long-duration timing note: the 7/30/90/365-day schedules must not be loaded blindly as very large constants into a standard timer. Confirm the FX5U timer instruction and supported preset range; use RTC/calendar comparison, a daily counter, or cascaded timers/counters if the interval exceeds the selected timer range.

CHAPTER 5: RESULTS & DISCUSSION

5.1 Logic-Level Results

The supplied images show the proposed warning-rung layout, the four maintenance categories and a GX Works3 monitoring window. Based on the visible rung structure, the intended behavior is that the timer-done contact initiates the corresponding warning, the warning contact provides a seal-in path, and the normally closed reset contact clears that warning when maintenance is acknowledged.

Test case	Expected output / observation	Verification status
Weekly due flag activated	Y0 warning active; Y4 buzzer active.	Logic criterion defined; test after device map is reconciled.
Monthly due flag activated	Y1 warning active; Y4 buzzer active.	Logic criterion defined; test in simulation.
Two warnings active together	Both task warnings stay active; Y4 remains ON.	Check parallel buzzer branch for Y0-Y3.
Weekly reset pressed while only Y0 is active	Y0 clears; Y1-Y3 remain unaffected.	Verify reset is task-specific.
Reset a warning while another remains due	Reset warning clears; Y4 remains ON.	Verify OR logic and reset isolation.
PLC power interruption	Interval behavior resumes according to selected retentive / RTC design.	Requires hardware configuration test.

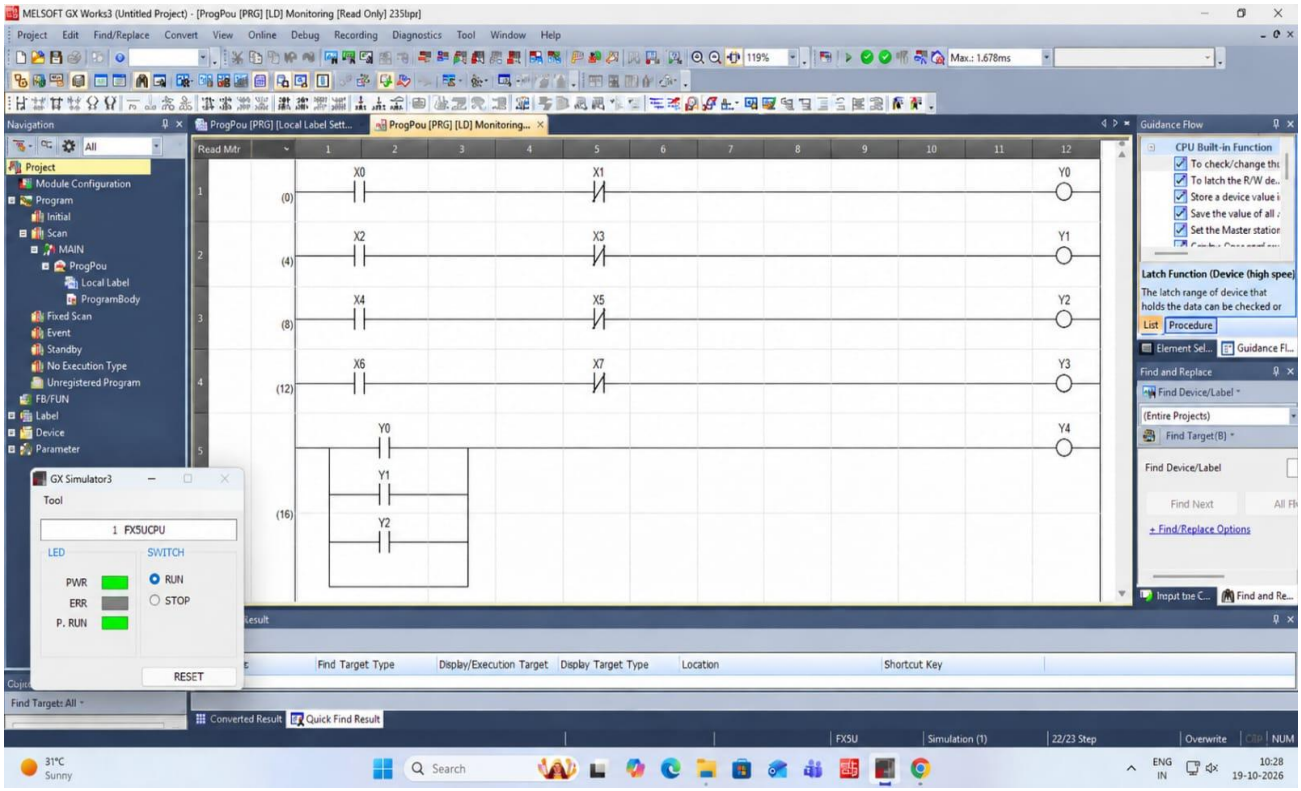
5.2 Discussion and Constraints

- The architecture is appropriate for a local maintenance reminder because task warning states are separated from the shared audible indication.
- The supplied GX Works3 screenshot is a snapshot of monitoring status. It does not provide evidence that the full weekly, monthly, quarterly or yearly durations have completed in real time.
- The two supplied ladder representations use different input-address conventions. Before commissioning, confirm a single I/O map, validate all reset contacts, include Y3 in the common buzzer branch, and update device comments.
- The program should use a valid extended time base or RTC/calendar comparison for long schedules; a basic timer's maximum preset may be too small for multi-day intervals.
- Field output types, contact voltage, buzzer current, isolation and fail-safe behavior must be checked against the exact PLC model and machine electrical drawings.

5.3 Acceptance Criteria

The design is ready for a controlled prototype test when each interval can be forced or simulated independently, each reset clears only its own warning, Y4 is ON whenever at least one warning is active, and all actual timer bases and retentive behaviors are documented. A technician should also confirm that the service task was physically completed before resetting the warning.

5.4 Output:


MELSOFT GX Works3 (Untitled Project) - [ProgPou [PRG] [LD] Monitoring [Read Only] 235bpr]
GX Simulator3
Tool
1 FX5UCPU
LED
SWITCH
PWR
ERR
P. RUN
RUN
STOP
RESET
X0
X1
Y0
X2
X3
Y1
X4
X5
Y2
X6
X7
Y3
Y0
Y1
Y2
Y4

CHAPTER 6: CONCLUSION

The PLC-Based Preventive Maintenance Warning System provides a straightforward ladder-logic concept for reminding maintenance staff about four recurring service activities: weekly lubrication and cleaning, monthly hydraulic-oil checking and filter replacement, quarterly pneumatic-system inspection, and yearly sensor calibration. Separate warning outputs Y0-Y3 identify each task, while a common buzzer output Y4 provides an audible alert whenever any warning remains active.

The report organizes the supplied diagrams and MELSOFT GX Works3 screenshot into a project design covering the maintenance schedule, system requirements, I/O allocation, program sequence, detailed rungs and logic-operation table. The key control principle is that a warning should remain active until the matching maintenance reset is operated, and resetting one category must not clear the others.

Before practical use, the design needs three important commissioning steps: (1) reconcile the different input-address conventions shown in the supplied images; (2) implement the extended weekly-to-yearly timing using a verified time base such as RTC/calendar comparison or cascaded counters; and (3) test simultaneous warnings, reset isolation, power recovery and electrical output ratings on the selected FX5U model.

Future Scope

- Add an HMI/SCADA panel showing each maintenance category, last reset date, next due date and alarm acknowledgement.
- Log the maintenance task, technician identity, date/time and downtime in a maintenance register or historian.
- Add password-protected reset permissions and a supervisor acknowledgement for overdue tasks.
- Use asset-specific maintenance intervals and calendar rules from the equipment manufacturer, including leap-year and month-end handling.
- Integrate the reminder with plant CMMS software after the local PLC logic is verified.

Safety and Use Note

The warning buzzer is a maintenance reminder and must not be treated as a safety function. It does not replace machine guards, emergency stops, lockout/tagout, hydraulic depressurization, pneumatic isolation, or the maintenance instructions supplied by the equipment manufacturer.

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2. Mitsubishi Electric Corporation, MELSOFT GX Works3 Operating Manual, applicable revision. Mitsubishi Electric Corporation.

3. W. Bolton, Programmable Logic Controllers, 6th ed., Newnes / Elsevier, 2015.

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7. Mitsubishi Electric Corporation, MELSEC iQ-F Series programming and instruction reference materials for timers, special relays and device addressing, applicable revision.

8. Project source material supplied for this report: GX Works3 ladder-monitor screenshot; PLC-based preventive-maintenance ladder overview; and detailed timer, warning and common-buzzer ladder illustration.

Note: Verify the revision/date of the Mitsubishi Electric manuals actually used in the laboratory and replace "applicable revision" with the bibliographic details available to the project team before final submission.